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Natural Language Processing

Session 1: Motivation & Course Overview

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Meet the teaching team

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Instructor: Luis Fernando Ramírez Ruiz

Doctoral candidate and researcher at the Robert Koch Institute

Research: machine learning + unstructured / multimodal data for climate-change and public-health narratives

- Hertie MPP + MDS
- Political Science (ITAM)
- Philosophy (UNAM)

Contact: l.ruiz@hertie-school.org

Teaching Assistant: Lonny Chen

The TA will lead the lab sessions and support the practical implementation of the lecture material.

Office-hour arrangements will be announced on Moodle.

- Hertie MDS, Research Assistant
- Computer Engineering (UBC)

Contact: l.chen@hertie-school.org

Course objectives, expectations & grading

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OBJECTIVES

1. Understand core NLP concepts, assumptions, and tools

2. Implement NLP applications in Python

3. Evaluate and communicate model performance

4. Build toward independent, realistic NLP systems

EXPECTATIONS

Come prepared: do the required reading

Participate actively and ask questions

Use the labs to turn concepts into working code

Prerequisites: Statistics II / causal inference + introductory Python or R

GRADING

10% Class participation

30% Best three out of four quizzes

20% Outline for research note

40% Research note

Research note: extend a published study OR pursue your own research question

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Who is in the room?

We will ask three quick questions:

1. **Which Hertie program are you in?**

   MPP · MDS · MIA · exchange / other

2. **What is your academic background?**

   Social sciences · economics · policy/law · data/CS · humanities · other

3. **What is your programming starting point?**

   Python · R · both · basic experience · very little / none



What do you expect from this course?

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RESEARCH

What kind of question or text data would you like to analyze?

SKILLS

What would you like to be able to build or evaluate?

CONCERNS

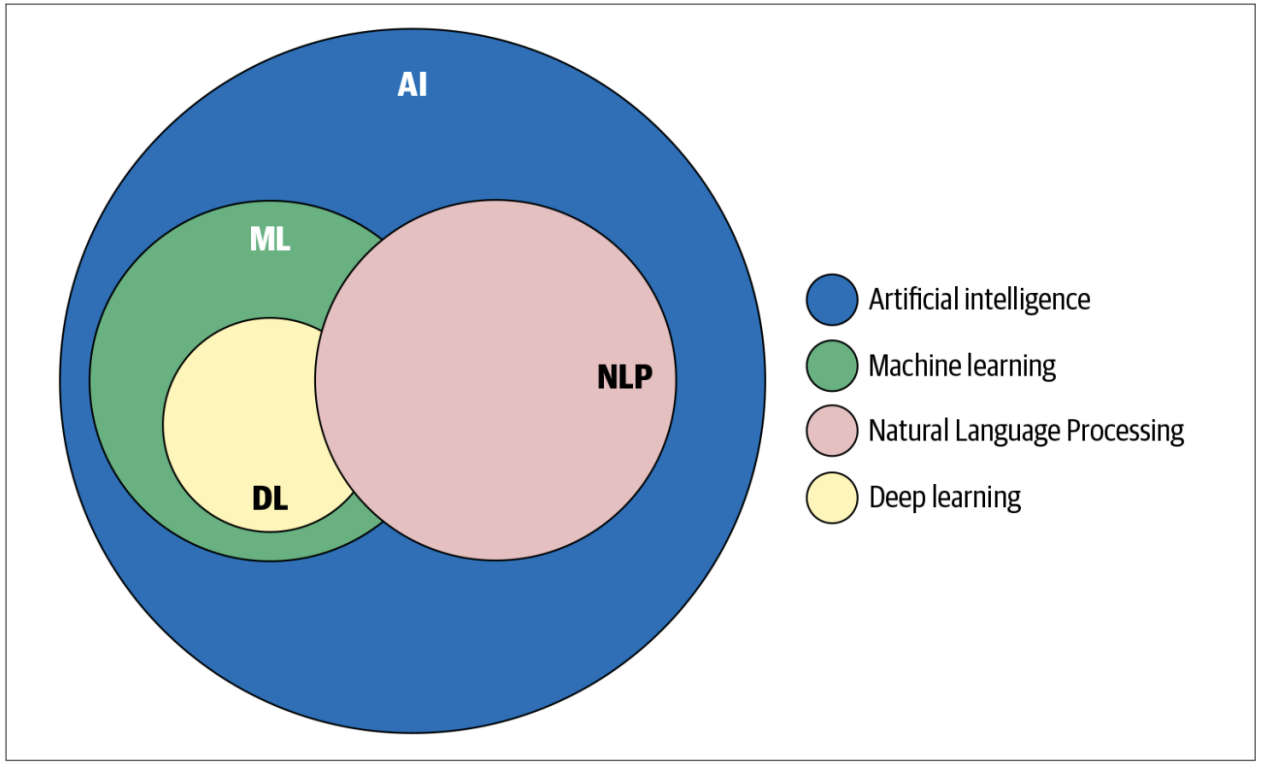
What currently feels most difficult or uncertain about NLP?

Session Break

When we return: What is NLP, where does it sit in AI / ML / deep learning, and where do we already encounter it?

Where does NLP sit in AI, machine learning, and deep learning?

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Source: Vajjala et al. (2020), ch. 1, p.15

NLP: LANGUAGE AS THE PROBLEM DOMAIN

Understanding, representing, retrieving, classifying, and generating language.

RULES

- patterns
- lexicons

ML

- features
- labels
- probabilistic models

DL / LLMs

- Learned Latent representations
- Transfer learning
- Fine-tuning models

Important: NLP is not simply a subset of deep learning. Modern NLP often uses ML/DL, but rule-based and hybrid systems remain useful.

NLP is already in your day

Note: Language processing is embedded in ordinary tools, often without looking like “AI-powered”.



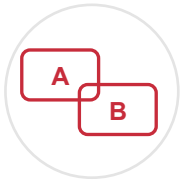
INBOX

filter · prioritize



SEARCH

query · retrieve



VOICE / TRANSLATE

speech · meaning



DOCUMENTS

extract · summarize



MESSAGING

suggest · rewrite

Same language, different tasks

NLP is usually one layer of a larger system

What do NLP systems actually do?

CLASSIFY ➡ choose a label or score

FIND / EXTRACT ➡ retrieve relevant text or facts

GENERATE ➡ produce language from context

Most real systems chain these operations together.



Four levels of language

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01	02	03	04
SOUND / SYMBOLS	**WORD FORM**	**STRUCTURE**	**MEANING IN CONTEXT**
phonemes & graphemes	**morphemes & lexemes**	**syntax**	**semantics + pragmatics**
Smallest spoken or written units.	Meaningful word parts and related forms.	How words combine and depend on one another.	What the sentence means — and what it means here.
	un + break + able	*who did what to whom?*	*“Great.” → praise or sarcasm*

more local information → more contextual information

Different tasks “look” at different levels of this information.

Exercise: where might an NLP system get confused?

What would a machine need in order to interpret each case correctly?

1. "I made her duck."

2. "The man couldn't lift his son because he was so weak."

3. "man bit dog" / "dog bit man"

2 min in pairs → 3 min discussion

How NLP methods become more expressive

RULES & HEURISTICS

Developers encode patterns using domain expertise, dictionaries, regexes, grammars, and other knowledge resources.

MACHINE LEARNING

The system learns a task from training examples after text is converted into numerical features; supervised learning uses labels.

DEEP LEARNING

Neural models learn stronger representations for complex, unstructured language; large transformers add pretraining and transfer.

WHAT VAJJALA EMPHASIZES

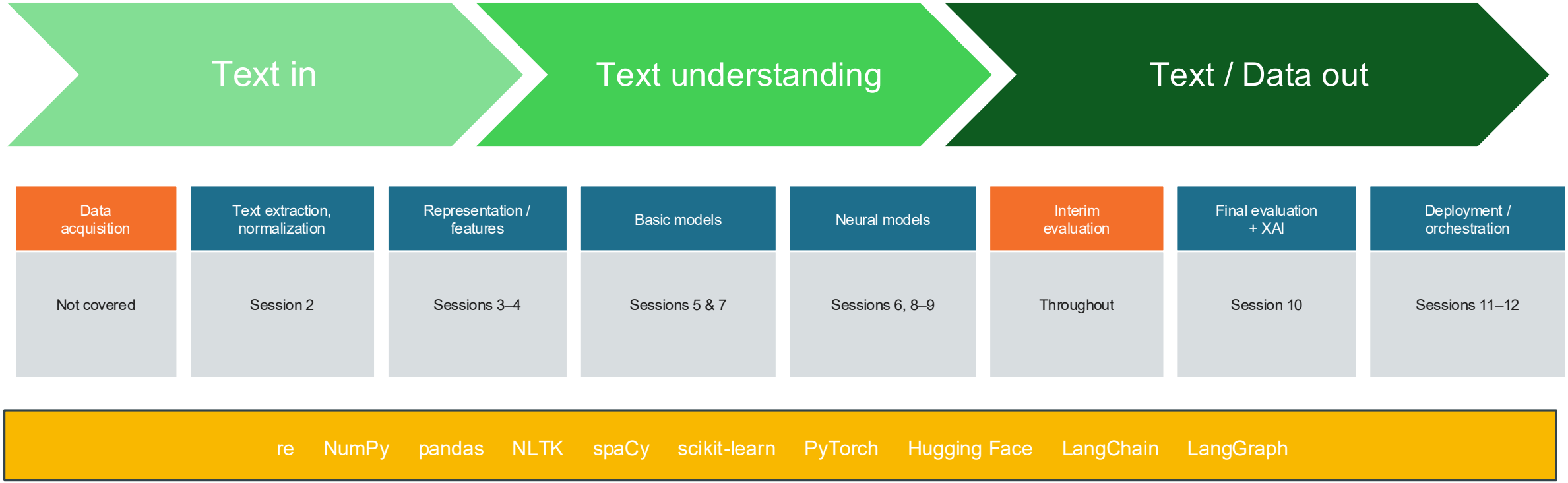
Rules are useful for quickly building a first version and for plugging costly gaps in production systems.

ML replaces **explicit instructions** with **patterns** learned from examples, but depends on the representation and training data provided.

Deep learning is motivated by language being “complex and unstructured”; transformers extend this with **large-scale pretraining and reuse across tasks**.

Key idea: greater representation and learning capacity expands what can be modeled; it does not make older methods obsolete.

The NLP pipeline



Conceptual (lecture) and applied (labs) perspective

From problem to model: three linked choices

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1 · PROBLEM FORMULATION

What is the unit?

- Message
- Sentence
- Document
- Conversation

What is the final target?

- label
- score
- ranking
- generated text

2 · REPRESENTATION

How do we make language computable?

- counts / TF–IDF
- static embeddings
- contextual embeddings
- structured features

What type of information is lost?

3 · MODEL CHOICE

- Is a rule-based model enough?
- Is Logistic regression doing a good job in identfying labels and categories?
- Do we need a neural classifier?
- Can this problem be solved with a Transformer or LLM?

NOTE:

- Choose the simplest model that meets the task, data, and deployment constraints.

A bigger model cannot rescue a poorly specified problem.

Keep this mental model

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LANGUAGE → REPRESENTATION → MODEL → EVALUATION

A useful NLP system starts with the problem — not with the fanciest model.

- NLP is broader than LLMs.
- Language has structure, ambiguity, and context.
- Every representation makes some information visible — and hides something else.

Next week → strings · regex · tokenization · preprocessing